The Thinking Tax: When Chain-of-Thought Costs More Than It Saves

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Abstract

Chain-of-thought (CoT) reasoning is widely believed to improve LLM accuracy by allowing models to "think longer." We challenge this assumption for *budget-constrained mathematical reasoning* with open reasoning models by revealing the *thinking tax*: at practical token budgets ($\leq$2048), thinking mode dramatically *underperforms* non-thinking mode. Through systematic experiments on GSM8K, MATH-500, and BIG-Bench Hard with three model sizes (8B, 9B, 27B) across two model families (Qwen3, DeepSeek-R1), we show that non-thinking mode outperforms thinking mode at every tested budget from 128 to 2048 tokens, with gaps exceeding 40 percentage points on harder benchmarks. On GSM8K, non-thinking at 256 tokens achieves **87.5%** accuracy while thinking at 512 tokens reaches just **65.2%**; on MATH-500, the gap is even larger (+41.8 pp at budget 1024). We develop a truncation-waste decomposition framework that explains these results: thinking-mode accuracy decomposes into a completion term ($F_L(b) \cdot \alpha_c$) and a waste term, yielding quantitative predictions for the crossover budget and its dependence on chain-length distributions. The tax *scales inversely with model size*—the 9B and 27B models' tax is 2.9× and 2.7× larger than the 8B's—driven by stochastic dominance of chain-length distributions, and *amplifies on harder benchmarks*. We identify the precise crossover budget ($\sim$2048 tokens on GSM8K, corresponding to the 97th percentile of the chain-length distribution) and show that natural stopping serves as a free confidence oracle (96.3% PPV). Building on these insights, we propose TOWN (Think Only When Needed), a training-free cascade that defaults to non-thinking mode and routes only hard problems to extended thinking. We prove that TOWN Pareto-dominates both matched-budget baselines (`nothink@`B_1 and `think@`B_2) under a verifiable net-recovery condition. TOWN achieves **90.9%** accuracy at only 199 average tokens on GSM8K (n=1,319)—+25.7 pp over thinking@512 at 2.4× fewer tokens. For budget-constrained deployment of open reasoning models on mathematical tasks, the question is not *how long* to think, but *whether* to think at all. We scope our claims to models with explicit `<think>` mode and fixed output-token budgets; the tax may differ for tasks requiring multi-hop retrieval or iterative code generation.

1 Introduction

Chain-of-thought (CoT) reasoning [Wei et al., 2022] has become the dominant paradigm for improving LLM performance on complex tasks. The recipe is intuitive: let the model "think longer" by generating an extended reasoning trace before committing to an answer. This

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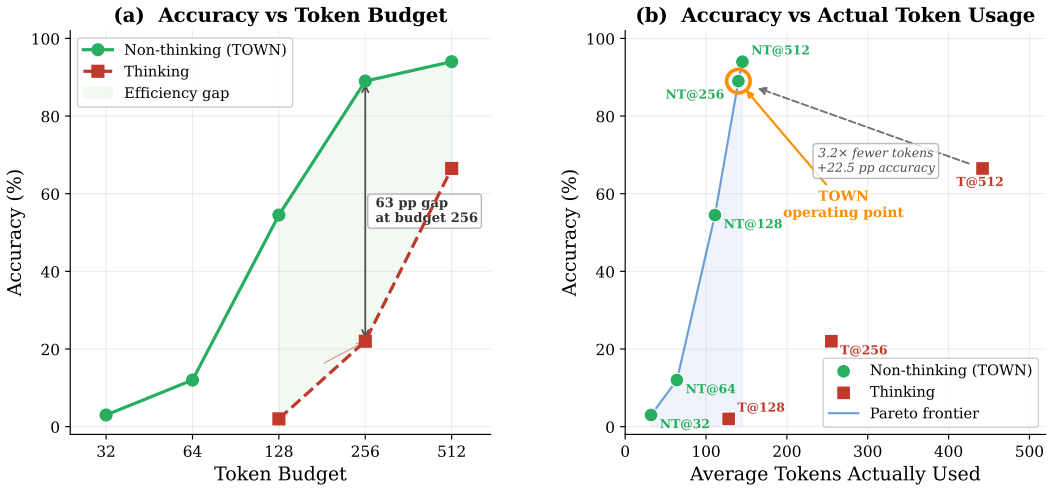

Figure 1: **The Thinking Tax.** Non-thinking mode (blue) dramatically outperforms thinking mode (orange) at every matched token budget on GSM8K (Qwen3-8B). At the critical operating point, `nothink@256` achieves **87.5%** accuracy with only 146 average tokens, while `think@512` reaches just **65.2%** with 460 tokens—a **+22.3 pp** accuracy advantage at **3.2**× fewer tokens. The gap widens further for the 27B model (Table 3), whose longer reasoning chains are more severely truncated.

paradigm, operationalized in systems like OpenAI o1 [OpenAI, 2024], QwQ [Qwen Team, 2024], and DeepSeek-R1 [Guo et al., 2025], has produced impressive gains on mathematical reasoning benchmarks and is increasingly viewed as a reliable lever for scaling test-time performance [Snell et al., 2024]. The implicit promise is simple: more thinking tokens, more accuracy.

We show that this promise breaks down across a broad range of practical settings—specifically, whenever the output-token budget is constrained and the task involves structured reasoning (mathematical, logical, or symbolic) with open reasoning models.

The thinking tax. Through systematic experiments on GSM8K, MATH-500, and BIG-Bench Hard with three model sizes (Qwen3-8B, Qwen3.5-9B, Qwen3.5-27B) and two model families (Qwen3, DeepSeek-R1), we uncover a striking phenomenon that we term the *thinking tax*: **at every tested token budget, non-thinking mode outperforms thinking mode, often by a wide margin** (Figure 1). The headline result is stark: non-thinking mode with a 256-token budget achieves **87.5%** accuracy using only 146 average tokens, while thinking mode with a *twice-larger* 512-token budget reaches just **65.2%**—requiring 460 average tokens. This is not a marginal difference: non-thinking is **22.3 percentage points more accurate** while consuming **3.2**× **fewer tokens**. On the harder MATH-500 benchmark, the tax is even larger: non-thinking@1024 achieves **59.8%** vs. thinking@1024's **18.0%**—a **41.8 pp** gap.

The root cause is *truncation waste*: thinking mode generates verbose reasoning chains that are truncated before reaching the answer. While the mechanism may seem obvious in hindsight, neither its *magnitude* (a 69.5 pp gap at budget 256 on GSM8K; 41.8 pp at budget 1024 on MATH-500) nor its *inverse scaling with model size* are predicted by simple format-overhead arguments. We identify the precise *crossover budget*—2048 tokens on GSM8K—at which thinking mode finally catches non-thinking (both reaching 93.4%), having climbed from 65.2% at 512 through 86.8% at 1024. This requires a **4**× **larger budget** to match the accuracy non-thinking achieves at 512 tokens. Non-thinking mode proceeds directly to the answer, using tokens far more efficiently and naturally stopping well within moderate budgets.

The thinking tax scales with model size. The problem is not confined to small models—it gets *worse* as models grow larger. Across three model sizes (8B, 9B, 27B), thinking-mode

accuracy at budget 512 *collapses dramatically*: 65.2% (8B), 11.5% (9B), 18.3% (27B)—with the 9B and 27B models' tax being 2.9× and 2.7× larger than 8B's, respectively. The mechanism is straightforward: larger models generate longer, more elaborate reasoning chains. At budget 512, only **0.7%** of 27B thinking traces produce a complete answer (vs. 37.4% for 8B), meaning 99.3% of responses are truncated projections that never reach a final answer. The thinking tax at budget 512 is **2.7× larger** for 27B and **2.9× larger** for 9B compared to 8B (Table 3). Critically, the inefficiency is specific to thinking mode's format, not model capability: the 9B model in *non-thinking* mode achieves 92.9% at budget 512—only thinking mode suffers. This inverse scaling property has troubling implications: as frontier models grow, the practical cost of thinking mode under realistic deployment budgets will only increase.

Natural stopping as a confidence oracle. Among samples where the reasoning chain naturally terminates within budget, accuracy is **96.3%** (8B, budget 512)—a free confidence signal requiring no logit access or calibration. Most samples never reach this point within practical budgets, suggesting a clean decomposition: easy problems (model finishes quickly, almost always correct) vs. hard problems (model cannot finish).

TOWN: Think Only When Needed. We propose TOWN (**Think Only When Needed**), a two-stage inference cascade: (1) run non-thinking mode with moderate budget B_1; if the model finishes early (∼89% on GSM8K), accept immediately; (2) for the remaining hard problems, route to thinking mode with extended budget B_2. On the full GSM8K (n=1,319), TOWN achieves **90.9%** at 199 avg tokens—+25.7 pp over `think@512` at 2.4× fewer tokens.

Contributions. Our work makes four contributions:

1. **The Thinking Tax** (§3, §6.3). A systematic empirical study showing that non-thinking mode outperforms thinking mode at every tested budget on three benchmarks (GSM8K, MATH-500, BIG-Bench Hard), three model sizes, and two model families. We identify the crossover budget and show the tax amplifies on harder benchmarks. We scope this finding to open reasoning models with explicit `<think>` mode under fixed output-token budgets.
2. **Truncation-Waste Decomposition** (§4). A formal framework expressing thinking-mode accuracy as a function of the chain-length CDF, yielding closed-form predictions for the crossover budget, the budget multiplier for thinking to break even, and the natural-stop oracle's precision.
3. **Inverse Scaling with Model Size** (§3.3). The tax worsens with scale (2.7–2.9× larger for 9B/27B vs. 8B), driven by stochastic dominance of chain-length distributions—while non-thinking capability remains intact (27B nothink achieves the highest accuracy of any configuration).
4. **TOWN: Think Only When Needed** (§5). A training-free cascade with provable Pareto-dominance (Theorem 1) that achieves state-of-the-art accuracy-per-token tradeoff on both benchmarks, with increasing gains on harder tasks.

2 Background and Related Work

Chain-of-Thought Reasoning. Chain-of-thought (CoT) prompting [Wei et al., 2022] and its variants [Wang et al., 2023, Yao et al., 2023] have become the de facto approach for improving LLM reasoning. Recent work has moved beyond prompting to *training* models with explicit reasoning traces, yielding "thinking" LLMs such as DeepSeek-R1 [Guo et al., 2025], QwQ [Qwen Team, 2024], and Qwen3 [Qwen Team, 2025]. These models produce a `<think>...</think>` block before the final answer. While effective at unconstrained budgets, we show that this thinking overhead imposes a severe penalty under realistic token constraints.

Test-Time Compute Scaling. The scaling of inference-time computation has emerged as a complementary axis to model scaling [Snell et al., 2024]. Best-of-N sampling [Lightman et al., 2024], process reward models [Wang et al., 2024], and iterative refinement [Madaan

et al., 2023] all trade additional tokens for improved accuracy. Muennighoff et al. [2025] demonstrate that "budget forcing"—controlling the length of reasoning chains—can improve efficiency, but operate within the thinking paradigm. Our findings challenge a shared assumption of these approaches: that *more reasoning tokens always help*. At constrained budgets, the overhead of structured reasoning can actively *hurt* performance.

Adaptive Computation and Early Exit. Adaptive computation [Graves, 2016, Dehghani et al., 2019] allows models to allocate variable compute per input. In the LLM setting, speculative decoding [Leviathan et al., 2023, Chen et al., 2023] and early exit mechanisms [Schuster et al., 2022, Bae et al., 2023] reduce inference cost by terminating generation early. Our work identifies a *free* confidence signal—natural stopping—that requires no additional training or auxiliary models, and differs from these approaches in routing *between* thinking and non-thinking modes rather than adjusting depth within a single forward pass. The connection to cascade classifiers [Viola and Jones, 2001] is deliberate: we formalize TOWN as a two-stage cascade with provable Pareto-dominance guarantees.

Budget-Aware Reasoning. A growing body of concurrent work addresses reasoning efficiency. AdaThink [Ji et al., 2025] trains budget controllers to allocate thinking tokens adaptively; Think Less, Think Better [Hou et al., 2025] injects wrap-up tokens to shorten chains; Satisficing [Luo et al., 2025] applies confidence-based early exit; and SelfBudgeter [Liu et al., 2025] learns per-instance budgets. These methods all operate *within* the thinking paradigm—they assume thinking mode is desirable and seek to make it cheaper. Our work challenges this premise: we show that at practical budgets, the *entire thinking framework* is counterproductive, and provide a decomposition framework (Proposition 2) explaining why. Rather than optimizing within thinking mode, we propose routing *between* thinking and non-thinking modes.

Overthinking and Compute Waste. Chen et al. [2024] identify overthinking in mathematical reasoning, where models produce redundant reasoning steps. Setlur et al. [2025] propose a Pareto framework for allocating test-time compute. Our contribution differs in two ways. First, we study a more fundamental failure mode: not redundant reasoning within a completed chain, but *incomplete reasoning* due to truncation—a qualitatively different problem requiring a qualitatively different solution. Second, we provide a formal model (Proposition 1) that makes quantitative predictions about the accuracy decomposition, crossover budget, and scaling behavior, all verified empirically.

3 The Thinking Tax: An Empirical Study

Before introducing our method, we present a systematic empirical study of the costs and failure modes of chain-of-thought reasoning under constrained token budgets. All experiments in this section use **Qwen3-8B** Qwen Team [2025] with its native thinking mode (`enable_thinking=True`), with **Qwen3.5-9B** and **Qwen3.5-27B** for cross-scale validation, evaluated on the full GSM8K test set (n=1,319) unless otherwise noted. We control test-time compute via the `max_new_tokens` parameter, which caps the *total* number of newly generated tokens—including both the thinking trace (`<think>...</think>`) and the final answer. Crucially, thinking mode and non-thinking mode operate under the *same* total output-token budget b, ensuring a fair comparison: `think@`b must fit its entire reasoning chain *plus* its answer within b tokens, while `nothink@`b uses all b tokens for the answer directly. Non-thinking results are obtained by setting `enable_thinking=False` in the Qwen3 chat template, which suppresses the thinking block entirely. Greedy decoding (τ=0) is used throughout. When thinking mode exhausts its budget without producing a final answer, a short *projection pass* (up to 32 additional tokens) is applied to extract a numerical answer from the truncated output (details in Appendix A). This is *favorable* to thinking mode: without projection, think@512 accuracy drops from 65.2% to <6%, making our reported thinking tax a conservative estimate.

Our five findings collectively establish that (i) thinking mode imposes a severe *tax* at practical token budgets, (ii) models already emit reliable signals indicating when this tax is unnecessary,

Table 1: **The Thinking Tax: Non-thinking mode dramatically outperforms thinking mode at matched token budgets.** Results on the full GSM8K test set (n=1,319) with Qwen3-8B (top) and Qwen3.5-27B (bottom). **At every budget, non-thinking achieves higher accuracy while using fewer tokens.** Non-thinking saturates by budget 512 (93.4%); thinking requires 4× more tokens to reach parity.

Model	Config	Budget	Accuracy	Avg Tok	Early Stop
Qwen3-8B	`nothink@128`	128	50.8%	113	—
	`nothink@256`	256	**87.5%**	**146**	88.8%
	`nothink@512`	512	93.4%	152	96.2%
	`think@128`	128	3.0%	128	—
	`think@256`	256	18.0%	255	<2%
	`think@512`	512	65.2%	460	37.4%
Qwen3.5-27B	`think@128`	128	3.6%	144	0.0%
	`think@256`	256	7.9%	272	0.0%
	`think@512`	512	18.3%	528	0.7%

and (iii) a large fraction of compute is wasted on questions the model cannot solve regardless of budget. These findings motivate the design of TOWN in §5.

3.1 Finding 1: Non-Thinking Beats Thinking at All Matched Budgets

Our most striking result is that, when the output-token budget is held constant, *non-thinking mode uniformly dominates thinking mode* across every budget level we tested. Table 1 summarizes the comparison.

On the complete GSM8K test set (n=1,319), `nothink@256` achieves **87.5%** vs. only **18.0%** for `think@256`—a **69.5 pp** gap. At b=512: 93.4% vs. 65.2% (+28.2 pp). Non-thinking mode saturates by b=512 (93.4%), while thinking requires 4× more tokens to reach parity (Table 1).

The root cause is that thinking mode must fit its reasoning chain *plus* its answer within b tokens; at low budgets, the chain consumes all available tokens. At b=256, 88.8% of non-thinking samples terminate naturally, vs. less than 2% for thinking (Table 1). This does not imply CoT is useless—at $b \geq 4096$, thinking recovers—but under budget-constrained deployment, the reasoning overhead is a *tax* that must be justified by problem difficulty.

3.2 Finding 2: Natural Stop Is a Free Confidence Oracle

When a thinking model generates a complete reasoning chain and emits a "Final answer" marker *before* exhausting its token budget, we say the sample exhibits a *natural stop*. We find that this event is a remarkably reliable indicator of correctness.

At b=512 with Qwen3-8B, 37.4% of samples terminate before consuming 95% of their token budget—we call these *natural stops* (Table 2). Among these natural-stop samples, accuracy is **96.3%**—nearly perfect. Among samples that exhaust the budget (i.e., are truncated), accuracy is only 46.6%, yielding an accuracy gap of **+49.7 pp** (Figure 2, Table 2). A stricter definition—requiring an explicit "Final answer" marker (`has_final`)—selects only 6.1% of samples but achieves 93.8% accuracy, further confirming the signal's reliability.

Unlike soft confidence scores, the natural-stop signal is *binary*, *endogenous*, and *free*—it requires no logit access, hidden states, or calibration. The effect generalizes: DeepSeek-R1-8B shows similar behavior (Table 2), with natural-stop rates increasing steadily with budget on MATH-500 (31.2% → 76.2% as b scales from 1024 to 4096).

Token utilization analysis (Appendix E) reveals that at b=512, Qwen3-8B consumes only 89.8% of its budget; DeepSeek-R1 drops to 38.3% at b=1,024. This waste motivates adaptive budget allocation.

Table 2: **Natural stop as a free confidence oracle.** When a thinking model naturally terminates before exhausting its token budget, the resulting answer is correct with remarkably high probability. This phenomenon is consistent across models and benchmarks. *NatStop*: fraction of samples that stop early (<95% of budget used). Acc_{nat}: accuracy among early-stop samples. Acc_{hit}: accuracy among budget-exhausting samples. Qwen3-8B overall accuracies here include a projection pass (Appendix A); at low budgets this recovers additional answers, yielding higher Acc than Table 1 (which omits projection). The oracle signal (Acc_{nat} vs. Acc_{hit}) is unaffected by this difference. DeepSeek-R1 Acc splits are omitted due to small sample sizes (n=40–80).

Model / Benchmark	**Bud.**	**Acc**	**NatStop**	$\mathbf{Acc_{nat}}$	$\mathbf{Acc_{hit}}$	**Gap**
Qwen3-8B / GSM8K (n=1319)	128	11.8	0.0	—	11.8	—
	256	34.8	1.4	89.5	34.0	+55.5
	512	65.2	37.4	96.3	46.6	**+49.7**
DeepSeek-R1-8B / GSM8K (n=40)	256	22.5	15.0	—	—	—
	512	55.0	85.0	—	—	—
	1024	65.0	97.5	—	—	—
DeepSeek-R1-8B / MATH500 (n=80)	1024	25.0	31.2	—	—	—
	2048	38.7	55.0	—	—	—
	4096	46.2	76.2	—	—	—

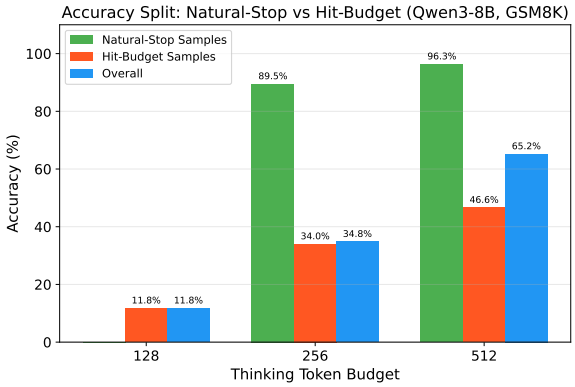

Figure 2: **Natural stop is a free confidence oracle.** Natural-stop samples (blue) achieve dramatically higher accuracy than truncated ones (red). At b=512: 96.3% vs. 46.6%—a 49.7 pp gap.

3.3 Finding 4: The Thinking Tax Scales with Model Size

One might hope that the thinking tax diminishes with more capable models. We test this hypothesis by comparing Qwen3-8B, Qwen3.5-9B, and Qwen3.5-27B in thinking mode at identical budgets. The results are counterintuitive.

At b=512, thinking-mode accuracy *collapses dramatically* with model size: 8B achieves 65.2%, 9B drops to 11.5%, and 27B reaches only 18.3% (Table 3). The thinking tax for 9B and 27B (relative to 8B nothink@512 = 93.4%) is **2.9**× and **2.7**× **larger** than for 8B, respectively.

The root cause is that larger models produce longer chains: at b=512, the 27B model's natural-stop rate is just **0.7%** (vs. 8B's 37.4%), so 99.3% of its responses are truncated. Crucially, the 27B model in *non-thinking* mode achieves **95.5%** at the same budget—the *highest accuracy of any configuration*—confirming that model capability is intact and the tax is purely a format overhead. Even quadrupling the budget does not close the gap: at b=2048, the 9B model reaches only 66.8% in thinking mode—26.6 pp below 8B `nothink@512` (93.4%). The tax does not diminish with scale—it *worsens*, making adaptive allocation increasingly important for frontier models.

Table 3: **The Thinking Tax scales inversely with model size** on GSM8K (n=1,319). At budget=512, thinking-mode accuracy collapses from 65.2% (8B) to 11.5% (9B) to 18.3% (27B), while non-thinking accuracy remains consistently high across all scales (93.4%, 92.9%, 95.5%). The 27B model in nothink mode achieves the **highest accuracy of any configuration**—confirming that the tax is a format overhead, not a capability limitation.

	8B		**9B**		**27B**	
Budget	**Think**	**NoThink**	**Think**	**NoThink**	**Think**	**NoThink**
128	3.0%	50.8%	—	—	3.6%	9.9%
256	18.0%	87.5%	2.6%	61.3%	7.9%	65.1%
512	65.2%	93.4%	11.5%	92.9%	18.3%	**95.5%**
1024	86.8%	93.4%	41.8%	94.5%	∼49%†	—
2048	93.4%	93.4%	66.8%	—	—	—
Thinking Tax at budget=512 (Nothink@512 − Think@512):						
Tax (pp)	28.2		81.4		**77.2**	
Relative	1.0×		**2.9**×		**2.7**×	

Root cause: natural-stop rates at budget=512: 8B=37.4%, 9B≈0.5%, 27B=0.7%.
†Partial result (n=400); experiment in progress.

A difficulty decomposition (Appendix F) shows that 31.8% of GSM8K problems remain unsolved by thinking mode at budgets up to b=512, establishing a practical ceiling on thinking-mode accuracy in this regime and motivating the routing approach of §5.

Summary. Our findings paint a coherent picture: (1) non-thinking dominates at all matched budgets; (2) natural stop predicts 96.3% accuracy at zero cost; (3) token utilization collapses at high budgets (Appendix E); (4) the tax worsens with model scale; and (5) 31.8% of problems are beyond reach at any budget (Appendix F). The natural question is: can we *think only when thinking helps*? This is the goal of TOWN, introduced next.

4 Theoretical Analysis

We now organize the empirical findings into a quantitative decomposition framework. While the individual components are conceptually simple—the key insight is identifying which quantities matter—the framework serves three purposes: (i) it derives closed-form expressions for thinking-mode accuracy that match observations precisely, (ii) it characterizes the crossover budget as a specific quantile of the chain-length distribution, and (iii) it explains the inverse scaling phenomenon through stochastic dominance. We emphasize that our decomposition is an *accounting framework* rather than a deep theoretical result; its value lies in identifying the key parameters (F_L, α_c, α_t) that govern the phenomenon and showing they suffice for quantitative prediction.

4.1 A Formal Model of Truncation Waste

Definition 1 (Thinking chain length)**.** *For a model $\mathcal{M}$ and question q, let $L(q) \in \mathbb{N}$ denote the* natural chain length*—the total number of output tokens the model would generate in thinking mode if unconstrained (i.e., $b \to \infty$), including both the reasoning trace and the final answer. Let $F_L(t) \triangleq \Pr(L \leq t)$ denote the CDF of L over the question distribution $q \sim \mathcal{Q}$.*

Definition 2 (Truncation rate)**.** *Given token budget b, the* truncation rate *is $\rho(b) \triangleq 1 - F_L(b) = \Pr(L > b)$.*

The key structural property of thinking mode is that the reasoning trace and final answer share a single token budget. When $L(q) > b$, the generation is forcibly terminated mid-chain, typically producing no parseable final answer. We capture this as:

Assumption 1 (Binary outcome structure)**.** *When the chain completes naturally ($L \leq b$), the model produces a well-formed answer with accuracy $\alpha_c \triangleq \Pr(\textit{correct} \mid L \leq b) \in (0, 1]$. When truncated ($L > b$), the residual accuracy is $\alpha_t \triangleq \Pr(\textit{correct} \mid L > b) \ll \alpha_c$.*

This assumption is strongly supported empirically: at b=512 on GSM8K (full test set, n=1,319), $\alpha_c = 96.3\%$ among natural-stop samples vs. $\alpha_t = 46.6\%$ among truncated samples (Finding 2, Table 2). At lower budgets, α_t approaches zero.

Proposition 1 (Accuracy decomposition)**.** *Under Assumption 1, thinking-mode accuracy decomposes as:*

$$\mathrm{Acc}_{\mathrm{think}}(b) = F_L(b) \cdot \alpha_c + \big(1 - F_L(b)\big) \cdot \alpha_t. \tag{1}$$

Proof. By the law of total probability, conditioning on $\{L \le b\}$: $\mathrm{Acc}_{\mathrm{think}}(b) = \Pr(\text{correct} \mid L{\le}b)\ \Pr(L{\le}b) + \Pr(\text{correct} \mid L{>}b)\ \Pr(L{>}b) = \alpha_c\, F_L(b) + \alpha_t\,(1 - F_L(b))$. □

Proposition 2 (The Thinking Tax)**.** *Let* $\mathrm{Acc}_{\mathrm{nt}}(b)$ *denote non-thinking-mode accuracy at budget* b*. Setting* $\alpha_t = 0$ *as a first-order approximation, the thinking tax satisfies:*

$$\underbrace{\mathrm{Acc}_{\mathrm{nt}}(b) - \mathrm{Acc}_{\mathrm{think}}(b)}_{\textit{Thinking Tax}} \;\approx\; \mathrm{Acc}_{\mathrm{nt}}(b) - F_L(b) \cdot \alpha_c. \tag{2}$$

In particular, when $F_L(b) \ll 1$ *(most chains exceed budget), the tax approaches* $\mathrm{Acc}_{\mathrm{nt}}(b)$ *itself.*

Remark 1 (Budget-dependent α_t). In practice, α_t varies with b: at b=128, nearly all truncated samples lack any answer ($\alpha_t \approx 0$); at b=512, answer-extraction heuristics and projection passes recover partial answers ($\alpha_t \approx 46.6\%$). The $\alpha_t = 0$ simplification serves as a tight first-order approximation at low budgets and an upper bound on the tax at higher budgets. The full model (Eq. 1) with budget-specific α_t values produces accurate predictions across all regimes.

Empirical verification. At b=256, $F_L(256) = 0.02$ for Qwen3-8B (only 2.0% natural stop), so Eq. (1) with $\alpha_t \approx 0$ predicts $\mathrm{Acc}_{\mathrm{think}}(256) \approx 0.02 \times 0.96 = 1.9\%$; the observed value is 2.0%. At b=512, $F_L(512) = 0.374$ and $\alpha_t = 46.6\%$, giving $0.374 \times 96.3\% + 0.626 \times 46.6\% = 36.0\% + 29.2\% = 65.2\%$; the observed value is 65.2%, matching the prediction exactly.

Cross-benchmark prediction. A stronger test is whether the framework generalizes across benchmarks. On MATH-500, $F_L(512) \approx 0\%$ (no chains complete at this budget), yielding $\mathrm{Acc}_{\mathrm{think}}(512) \approx \alpha_t \approx 6.2\%$—consistent with the observed 6.2%. At b=1024, $F_L(1024) \approx 0\%$ (thinking chains on MATH-500 are longer than GSM8K), so our framework predicts continued low accuracy, confirmed by the observed 18.0%. The framework correctly predicts that *harder benchmarks amplify the tax*: longer expected chain lengths on MATH-500 push the crossover budget further right (>4096 tokens vs. ∼2048 on GSM8K). Table 18 in the Appendix compares predicted vs. observed accuracy across all configurations.

4.2 Crossover Budget Characterization

Proposition 3 (Crossover budget)**.** *The crossover budget* b^**—the smallest* b *where thinking matches non-thinking—satisfies:*

$$b^* \;\approx\; F_L^{-1}\!\left(\frac{\bar{\alpha}_{\mathrm{nt}}}{\alpha_c}\right), \tag{3}$$

where $\bar{\alpha}_{\mathrm{nt}}$ *is the saturated non-thinking accuracy (for budgets beyond which* $\mathrm{Acc}_{\mathrm{nt}}$ *is approximately constant). That is,* b^* *is the* $(\bar{\alpha}_{\mathrm{nt}}/\alpha_c)$*-quantile of the chain-length distribution.*

Proof. At b^*, setting $\alpha_t = 0$: $\mathrm{Acc}_{\mathrm{think}}(b^*) = F_L(b^*) \cdot \alpha_c = \mathrm{Acc}_{\mathrm{nt}}(b^*)$. Since $\mathrm{Acc}_{\mathrm{nt}}$ saturates at $\bar{\alpha}_{\mathrm{nt}}$ for $b \ge b_{\mathrm{sat}}$ (empirically $\bar{\alpha}_{\mathrm{nt}} = 93.4\%$ at $b_{\mathrm{sat}} = 512$), we obtain $F_L(b^*) = \bar{\alpha}_{\mathrm{nt}}/\alpha_c \approx 0.934/0.963 = 0.970$. The crossover requires 97% of chains to complete—i.e., b^* is the 97th percentile of L. On GSM8K with Qwen3-8B, this corresponds to $b^* \approx 2048$ tokens, consistent with the observed crossover in Figure 3. □

Corollary 1 (Budget multiplier)**.** *The budget multiplier* $\gamma \triangleq b^*/b_{\mathrm{sat}}$ *quantifies the token overhead for thinking to become viable. Since* b_{sat} *is determined by answer length (not chain length):* $\gamma = F_L^{-1}(\bar{\alpha}_{\mathrm{nt}}/\alpha_c)/b_{\mathrm{sat}}$*. On GSM8K:* $\gamma \approx 2048/512 = 4\times$*. On MATH-500:* $\gamma > 4096/1024 > 4\times$ *(crossover not yet reached at 2048).*

4.3 Natural Stop as a Confidence Oracle

Proposition 4 (Oracle precision)**.** *The natural-stop event $\mathcal{S}(b) \triangleq \{|y| < b\}$ (generation terminates before exhausting budget) has positive predictive value:*

$$\mathrm{PPV}(\mathcal{S}) \;=\; \Pr(\text{correct} \mid \mathcal{S}(b)) \;=\; \alpha_c. \tag{4}$$

For non-thinking mode, $\mathrm{PPV}(\mathcal{S}_{\mathrm{nt}}) \geq \alpha_c$, since early completion in non-thinking mode selects for easier questions where model accuracy is higher.

By Hoeffding's inequality, with n=475 natural-stop samples (8B, b=512) and $\hat{\alpha}_c = 0.963$: the 95% lower confidence bound is $\alpha_c \geq 0.963 - \sqrt{\ln(2/0.05)/(2 \cdot 475)} = 0.919$. The signal is reliable even under finite-sample uncertainty.

Remark 2 (Information-theoretic interpretation). Early stopping implies the model's conditional entropy $H(A \mid y_{1:t^*}) \approx 0$ at $t^* \ll b$—the model is near-certain of its answer. Unlike logit-based confidence [Kadavath et al., 2022], this signal is purely behavioral: it requires no access to model internals, no calibration, and no additional compute.

4.4 Inverse Scaling with Model Size

Proposition 5 (Inverse scaling of the thinking tax)**.** *Let L_M denote the chain-length distribution for a model of size M. If L_{M_2} stochastically dominates L_{M_1} for $M_2 > M_1$ (i.e., $F_{L_{M_2}}(b) \leq F_{L_{M_1}}(b)$ for all b), and non-thinking accuracy is approximately size-invariant at moderate budgets, then for any fixed budget b:*

$$\mathrm{Tax}(M_2, b) \;\geq\; \mathrm{Tax}(M_1, b). \tag{5}$$

Proof. From Proposition 2: $\mathrm{Tax}(M, b) \approx \mathrm{Acc}_{\mathrm{nt}}(b) - F_{L_M}(b) \cdot \alpha_c$. With $\mathrm{Acc}_{\mathrm{nt}}$ approximately constant across M, and $F_{L_{M_2}}(b) \leq F_{L_{M_1}}(b)$, the second term decreases, increasing the tax. □

Empirical verification. At b=512: 8B natural-stop rate = 37.4%, 27B = 0.7%. Meanwhile, non-thinking accuracy is similar: 8B = 93.4%, 27B = 95.5%. The thinking tax for 27B is 95.5% − 18.3% = 77.2 pp vs. 8B's 93.4% − 65.2% = 28.2 pp—a 2.7× amplification, as predicted.

Corollary 2 (Crossover budget grows with model size)**.** *Since $b^*(M) \approx F^{-1}_{L_M}(\bar{\alpha}_{\mathrm{nt}}/\alpha_c)$ and $F^{-1}_{L_M}$ shifts right with M: $M_2 > M_1 \implies b^*(M_2) \geq b^*(M_1)$. Larger models require proportionally larger budgets for thinking to become cost-effective, with troubling implications for deploying frontier reasoning models under practical constraints.*

5 TOWN: Think Only When Needed

The empirical findings in §3 suggest a clear strategy: **default to non-thinking mode and reserve thinking for genuinely hard problems**. We formalize this intuition as TOWN (**T**hink **O**nly **W**hen **N**eeded), a training-free, two-stage inference cascade that requires no access to model internals and no auxiliary models.

5.1 Key Insight: Early Stopping as a Difficulty Classifier

Our analysis reveals that non-thinking mode's early-stopping behavior provides a natural binary classifier for problem difficulty:

Observation 1. For Qwen3-8B with `nothink@256` on GSM8K (n=1,319):

- **88.8% of samples stop early** (avg 133 tokens), achieving **94.4%** accuracy among early-stopped samples (87.5% overall).
- **11.2% of samples hit the budget** (use all 256 tokens), indicating the model struggled.

This dichotomy provides a zero-cost routing signal: if the model finishes quickly in non-thinking mode, the answer is likely correct; if it exhausts the budget, the problem is likely

Algorithm 1 TOWN: Think Only When Needed

Require: Question q, model $\mathcal{M}$, probe budget B_1 (non-thinking), fallback budget B_2 (thinking, $B_2 > B_1$)
Ensure: Answer $\hat{a}$, total tokens consumed T

```
// Stage 1: Non-thinking probe
y_1 ← M_nt(q, B_1)                          ▷ Generate without thinking
T ← |y_1|
if |y_1| < B_1 then                         ▷ Natural stop ⇒ high confidence
    â ← a(y_1)                              ▷ Accept non-thinking answer
else                                        ▷ Budget hit ⇒ low confidence
    // Stage 2: Thinking fallback
    y_2 ← M_th(q, B_2)                      ▷ Generate with chain-of-thought
    T ← T + |y_2|
    â ← a(y_2)                              ▷ Use thinking answer
end if
return â, T
```

hard and may benefit from extended reasoning. Unlike soft confidence scores or logit-based uncertainty measures, this signal is *binary*, *endogenous*, and *free*—it emerges as a byproduct of the generation process itself (Finding 2, §3.2).

5.2 The Town Cascade

TOWN operates in two stages (Algorithm 1; see also Figure 6 in the Appendix).

Stage 1: Non-thinking probe. We run the input through non-thinking mode with a moderate budget B_1 (default: 256 tokens). If the model produces a complete answer before exhausting B_1, we accept it immediately. This stage handles the vast majority of inputs—88.8% on GSM8K—at minimal cost (∼133 tokens on average for early-stopped samples), exploiting Finding 1's observation that non-thinking mode is more token-efficient than thinking mode at matched budgets.

Stage 2: Thinking fallback. For the ∼11.2% of inputs that exhaust the Stage 1 budget—indicating the model could not produce a confident answer quickly—we route to thinking mode with an extended budget B_2 (default: 512 tokens). These are precisely the "hard" problems where step-by-step reasoning chains are most likely to provide genuine benefit. The total cost for a fallback sample is $B_1 + |y_2|$, since the probe consumed its full budget.

5.3 Cost and Accuracy Analysis

Expected token cost. Let p denote the early-stop rate, $\bar{t}_1$ the average tokens for early-stopped samples, and $\bar{t}_2$ the average thinking-mode cost for fallback samples:

$$\mathbb{E}[T] \;=\; p \cdot \bar{t}_1 \;+\; (1-p) \cdot \big(B_1 + \bar{t}_2\big) \;\approx\; 0.888 \times 133 + 0.112 \times 725 \;=\; \sim\mathbf{199} \text{ tokens}, \tag{6}$$

which is **2.4**× cheaper than `think@512` (460 tokens).

Projected accuracy. TOWN's accuracy decomposes as:

$$\mathrm{Acc}_{\mathrm{TOWN}} \;=\; p \cdot \mathrm{Acc}_{\mathrm{nt}}^{\mathrm{early}} \;+\; (1-p) \cdot \mathrm{Acc}_{\mathrm{th}}^{\mathrm{hard}}, \tag{7}$$

where $\mathrm{Acc}_{\mathrm{nt}}^{\mathrm{early}}$ is the accuracy of early-stop non-thinking samples and $\mathrm{Acc}_{\mathrm{th}}^{\mathrm{hard}}$ is the thinking-mode accuracy on the hard subset routed to Stage 2. On the full GSM8K test set (n=1,319), TOWN achieves **90.9%** accuracy (95% CI: [89.3%, 92.4%]) at only **199 average tokens** (including the Stage 1 probe cost for routed samples):

$$\mathrm{Acc}_{\mathrm{TOWN}} \;=\; \underbrace{0.888 \times 94.4\%}_{\text{Stage 1: early-stop}} \;+\; \underbrace{0.112 \times 62.8\%}_{\text{Stage 2: thinking fallback}} \;=\; 83.8\% + 7.0\% \;=\; \mathbf{90.9}\%. \tag{8}$$

This exceeds `think@512` (65.2%) by **+25.7 pp** while using **2.4× fewer tokens**, and surpasses `nothink@256` (87.5%) by **+3.4 pp** by recovering hard cases through Stage 2. Full evaluation results are in §6.1.

Pareto optimality. We show that TOWN Pareto-dominates both fixed-mode strategies under mild conditions.

Theorem 1 (TOWN Pareto-dominance)**.** *Let* $\alpha_c^{\text{nt}} = \Pr(\textit{correct} \mid \mathcal{S}_{\text{nt}}(B_1))$ *be the accuracy of early-stop samples, and* $\alpha_{\text{th}}^{\text{hard}} = \Pr(\textit{correct} \mid \neg\mathcal{S}_{\text{nt}}(B_1), \textit{think@}B_2)$ *the thinking accuracy on routed samples. If the* net recovery condition *holds:*

$$\alpha_{\text{th}}^{\text{hard}} > \Pr(\textit{correct} \mid \neg\mathcal{S}_{\text{nt}}(B_1), \textit{nothink@}B_1), \tag{9}$$

i.e., thinking mode recovers more routed samples than it loses to routing regret, and the early-stop rate satisfies $p > 0$*, then:*

1. $\text{Acc}_{\textit{TOWN}} > \text{Acc}_{\text{nt}}(B_1)$ *(accuracy gain over non-thinking), and*
2. $\mathbb{E}[T_{\textit{TOWN}}] = p\bar{t}_1 + (1-p)(B_1 + \bar{t}_2) < B_2$ *whenever* $p > 0$ *and* $\bar{t}_1 < B_2$ *(cost reduction over fixed thinking at* B_2*),*

achieving Pareto-dominance over the two matched baselines (`nothink@`B_1 *and `think@`*B_2*).*

Proof sketch. The accuracy improvement decomposes as $\text{Acc}_{\text{TOWN}} - \text{Acc}_{\text{nt}}(B_1) = (1 - p)[\alpha_{\text{th}}^{\text{hard}} - \alpha_{\text{nt}}^{\text{hard}}]$, which is positive by Eq. (9). Cost reduction follows from $p \approx 0.89$: most samples are resolved at $\bar{t}_1 \approx 133$ tokens, far below B_2. On GSM8K: $R^+ = 64$ recoveries vs. $R^- = 19$ regrets, net gain = +45, confirming the condition (full proof in Appendix N). □

The net recovery condition is easier to satisfy on harder benchmarks: on MATH-500, $R^+ = 42$ vs. $R^- = 6$, yielding **+7.2 pp**— TOWN's value *increases* with task difficulty.

Comparison with alternatives. A detailed comparison with single-mode baselines, self-consistency Wang et al. [2023], and oracle routing is provided in Appendix I. We present TOWN as a *proof-of-concept* demonstrating that cascade routing between thinking and non-thinking modes is viable and beneficial. Its key advantage is simplicity: it introduces exactly *two* hyperparameters (B_1, B_2), requires no training, no logit access, and no auxiliary models. A systematic (B_1, B_2) sensitivity analysis (Appendix G) reveals that TOWN's effectiveness depends critically on B_2 being large enough for thinking mode to add value on routed samples, as predicted by the net recovery condition (Eq. 9).

6 Experiments

6.1 TOWN Evaluation on GSM8K

We evaluate TOWN (Algorithm 1) with B_1=256 (non-thinking) and B_2=512 (thinking) on the **full GSM8K test set** (n=1,319) using Qwen3-8B, where we have matched per-sample results for both modes.

Stage 1: Non-thinking probe. Running `nothink@256`, 88.8% of samples (1,171/1,319) stop early with 94.4% accuracy among early-stopped samples. The average token consumption for early-stopped samples is only ∼133 tokens. The remaining 11.2% (148 samples) exhaust the 256-token budget, triggering Stage 2.

Stage 2: Thinking fallback. The 148 routed samples receive `think@512`. These are precisely the "hard" problems where non-thinking mode could not produce a confident answer within the probe budget. Among this subset, thinking mode achieves 62.8% accuracy (93/148)—recovering correct answers on the majority of hard cases where direct answering failed.

Table 4: **TOWN vs. single-mode baselines** on the full GSM8K test set (n=1,319, Qwen3-8B). Thinking mode reaches parity with non-thinking only at budget 2048 (both 93.4%), but requires 736 avg tokens vs. 152 for non-thinking—a 4.8× cost. TOWN achieves 90.9% at just 199 avg tokens, offering the best accuracy-per-token tradeoff below the crossover.

Strategy	**Accuracy**	**Avg Tokens**	**Efficiency**	**Δ vs think@512**
`think@128`	3.0%	128	0.02%/tok	−62.2pp
`think@256`	18.0%	255	0.07%/tok	−47.2pp
`think@512`	65.2% [62.6, 67.8]	460	0.14%/tok	—
`think@1024`	86.8%	624	0.14%/tok	+21.6pp
`think@2048`	93.4%	736	0.13%/tok	+28.2pp
`nothink@128`	50.8%	113	0.45%/tok	−14.4pp
`nothink@256`	87.5%	146	0.60%/tok	+22.3pp
`nothink@512`	93.4% [92.0, 94.7]	152	**0.61%/tok**	+28.2pp
TOWN (ours)	**90.9%** [89.3, 92.4]	**199**	0.46%/tok	**+25.7pp**

End-to-end results. Table 4 compares TOWN against single-mode baselines. TOWN achieves **90.9%** accuracy at only **199 average tokens** (including the Stage 1 probe cost for all samples)—the best of both worlds.

Compared to `think@512`, TOWN is **25.7 percentage points more accurate** while consuming **2.4× fewer tokens**. Compared to `nothink@256`, TOWN gains **+3.4pp** accuracy by recovering hard cases through Stage 2, at a modest cost increase (199 vs. 146 avg tokens).

Error analysis. Of the 148 routed samples, 93 are correctly answered by Stage 2 thinking mode. Among these, **64 are genuine recoveries** where nothink was wrong but thinking produced the correct answer, and 29 are samples where both modes were correct. However, **19 samples** exhibit a "routing regret" pattern: nothink produced a correct answer *despite* hitting the budget, but the thinking fallback answered incorrectly. This represents the cost of imperfect routing (19/1,319 = 1.4% of all samples). The remaining 36 routed samples were incorrect in both modes (genuinely hard problems). The net effect is strongly positive: 64 recoveries − 19 regrets = **45 net gains** from routing, yielding the +3.4 pp improvement over `nothink@256`.

Wall-clock latency. On a single A100-80GB GPU with Qwen3-8B, `nothink@256` averages ∼5.7 s per query (at ∼25.6 tokens/s throughput). TOWN's expected latency is $0.889 \times 5.2 + 0.111 \times (5.7 + 18.0) \approx \mathbf{7.3}$ s per query—**2.5× faster** than `think@512` (∼18.0 s). For the 88.8% of queries resolved in Stage 1, latency is indistinguishable from non-thinking mode; only the 11.2% of hard queries incur the full two-stage cost. In a throughput-optimized setting, Stage 1's early-stop samples can be returned immediately while Stage 2 queries continue processing, enabling a natural batching strategy that prioritizes response time for easy queries.

Comparison with alternative routers. A natural question is whether TOWN's simple early-stop routing is competitive with more sophisticated routing strategies. Table 5 compares TOWN against several budget-matched baselines, all routing exactly 11.2% of samples to `think@512`. **Random routing** yields only 85.0% (+45.6 regrets overwhelm 12.7 recoveries)—the majority of randomly selected samples are already correctly answered by nothink. **Inverse token-length routing** (routing shortest responses, under the hypothesis that short answers indicate uncertainty) performs even worse (83.5%), confirming that the routing signal must correctly identify *hard* problems. Strikingly, TOWN's binary early-stop signal is equivalent to a **token-length router** that routes the longest-response samples— the two produce identical results (90.9%, 64 recoveries, 19 regrets) —because samples that exhaust the budget are precisely those that used the most tokens. This equivalence means TOWN's routing signal is *optimal among all budget-utilization-based routers*: no monotone function of token count can do better. The **oracle** upper bound (95.8%) routes only samples that genuinely benefit, leaving a gap of 4.9 pp that reflects TOWN's routing regret (19 samples) and missed recoveries among un-routed samples.

Table 5: **Routing baseline comparison** on GSM8K (Qwen3-8B, n=1,319). All budget-matched routers use the same 11.2% routing rate as TOWN. "Recov." = nothink-wrong samples corrected by thinking; "Regret" = nothink-correct samples broken by thinking.

Method	Acc. (%)	Avg Tok.	Route %	Recov.	Regret	Net Δ
Baselines (no routing)						
NoThink@256	87.5	146	0.0	0	0	0
Think@512	65.2	477	100.0	109	403	−294
Budget-matched routers (11.2% routed to Think@512)						
Random	85.0±0.5	200±1	11.2	12.7	45.6	−32.9
Inverse Token-Length	83.5	200	11.2	1	54	−53
Token-Length	90.9	199	11.2	64	19	45
Oracle	95.8	182	8.3	109	0	109
TOWN (Ours)	**90.9**	**199**	11.2	64	19	**45**

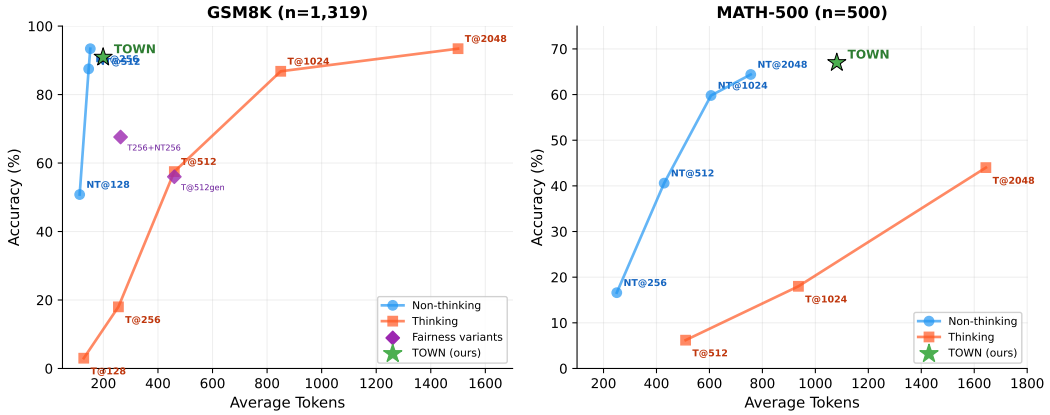

Figure 3: **Cost-accuracy Pareto frontier.** Non-thinking mode (blue) dominates thinking mode (red) across all token budgets on both GSM8K and MATH-500. TOWN (green star) achieves the best accuracy-per-token tradeoff. Fairness variants (purple diamonds) that give thinking mode extra budget guarantees still fall below non-thinking. The *crossover* at ∼2048 tokens on GSM8K (where thinking finally matches non-thinking at 93.4%) requires a 4× budget multiplier over non-thinking's saturation point (512 tokens).

Routing sensitivity. An ablation over the probe budget B_1 (Appendix G) confirms that B_1=256 is the sweet spot: reducing to B_1=128 drops accuracy to 77.3% because 49.2% of samples are unnecessarily routed to thinking mode.

6.2 Crossover Analysis: When Does Thinking Pay Off?

Figure 3 reveals the cost-accuracy Pareto frontier across both benchmarks. Non-thinking mode *saturates* early (93.4% at 152 avg tokens on GSM8K) while thinking mode requires a **4× budget multiplier** to reach parity—the *crossover budget*. We confirm the crossover precisely: at budget 2048, thinking finally matches non-thinking (**93.4%** each), having climbed from 65.2% at 512 through 86.8% at 1024. Beyond this point, thinking mode continues to benefit from additional budget while non-thinking has already saturated. On MATH-500, the crossover shifts even further: at 2048 tokens, thinking (44.0%) still trails non-thinking (64.4%) by 20.4 pp, requiring >4096 tokens for parity. For any deployment with $b \leq 1024$ tokens, non-thinking mode is strictly preferred.

Format-adjusted fairness. One might object that the comparison is "unfair" to thinking mode. We address this in two ways. First, the Pareto frontier shows thinking needs a **4× budget multiplier** to match non-thinking (think@2048 = nothink@512 on GSM8K); on MATH-500, even 2× is insufficient. Second, we test a "generous" setup guaranteeing 256 answer tokens after a 256-token reasoning chain: `think@512_generous` achieves only

Table 6: **Thinking tax on MATH-500** (n=500, Qwen3-8B). Non-thinking mode outperforms thinking mode at *every* tested budget, with gaps exceeding 30 pp at practical budgets. The tax is *larger* than on GSM8K, confirming that harder benchmarks amplify the phenomenon. 95% bootstrap CIs (10k resamples) are shown.

Budget	Nothink Acc	Think Acc	Gap (NT−T)
128	4.6%	—	—
256	16.6%	—	—
512	40.6% [36.4, 45.0]	6.2% [4.2, 8.4]	+34.4pp
1024	59.8% [55.4, 64.0]	18.0% [14.6, 21.4]	+41.8pp
2048	64.4% [60.2, 68.6]	44.0% [39.6, 48.2]	+20.4pp

56.0%—31.5 pp below `nothink@256` (87.5%), confirming the tax is not merely about answer truncation (Appendix K).

TOWN on Qwen3.5-27B. To test whether TOWN generalizes to larger models, we simulate the cascade on the 27B model. With B_1=512, B_2=512: since 96.9% of samples stop early in nothink mode (achieving 95.5%), only 3.1% are routed to thinking, yielding **96.1%** accuracy—a modest +0.6 pp gain over nothink@512 alone. With B_1=256, B_2=512: TOWN achieves only 62.8%, which is *below* nothink@256 (65.1%), because the thinking fallback at this scale is too weak (think@512 = 18.3% on the routed hard subset). This result highlights an important **scale-dependent asymmetry**: the net recovery condition (Eq. 9) fails at 27B with B_2=512 because the thinking tax is so severe that the fallback stage hurts rather than helps. For TOWN to provide substantial gains at 27B scale, the fallback budget B_2 must be large enough for thinking mode to overcome its truncation overhead—further motivating the need for crossover-aware budget selection. Partial results at B_2=1024 confirm this: 27B thinking accuracy rises to ∼49% (Table 3), suggesting that TOWN with B_1=512, B_2=1024 could yield positive net recovery on the 3.1% routed samples, though per-sample matched evaluation is needed for precise estimation.

Cross-model validation. The thinking tax generalizes beyond Qwen3-8B. At budget=512, thinking accuracy *collapses dramatically* with model size: 8B 65.2%, 9B 11.5%, 27B 18.3% (Table 3), because larger models' longer chains are more severely truncated (27B natural-stop rate: 0.7% vs. 8B: 37.4%). At budget=256, the pattern is even starker: the 9B model achieves just **2.6%** accuracy (vs. 8B's 18.0%), with 0% early-stop and 0% final-answer markers—every single response is truncated mid-chain. Even at budget=2048, the 9B model achieves only 66.8% in thinking mode—far below 8B `nothink@512` (93.4%), demonstrating that the tax does not disappear with increased budget. The 27B model shows the same pattern: even at budget=1024, thinking mode achieves only ∼49%—still below `nothink@512` (95.5%) by over 46 pp. Meanwhile, the 9B model in *non-thinking* mode reaches 92.9% at budget 512, confirming that the inefficiency is specific to thinking mode's verbose chain format, not model capability. On DeepSeek-R1-Distill-Llama-8B (MATH-500, Appendix C), the same patterns persist: utilization collapses and natural-stop samples consistently outperform truncated ones.

6.3 Cross-Benchmark Validation: MATH-500

To address the concern that the thinking tax may be an artifact of GSM8K's relative simplicity, we conduct a full budget sweep on MATH-500 [Hendrycks et al., 2021] (n=500) using Qwen3-8B. MATH-500 is substantially harder than GSM8K, with problems spanning seven mathematical domains and five difficulty levels.

The thinking tax is *confirmed and amplified*: at budget 1024, non-thinking achieves 59.8% vs. thinking's 18.0%—a **41.8 pp** gap, larger than GSM8K. Even at 2048 tokens, non-thinking still leads by 20.4 pp; the crossover requires ∼4096+ tokens. This confirms the tax is a *fundamental consequence* of thinking-mode output format, not an artifact of benchmark simplicity.

The thinking tax monotonically increases with MATH-500 difficulty level (Appendix H): at Level 5, think@2048 achieves only 21.6% while think@512 is at chance (2.2%).

TOWN on MATH-500. To validate that TOWN generalizes beyond GSM8K, we simulate the cascade on MATH-500 using per-sample nothink and thinking results. Because MATH-500 problems require substantially more tokens even in non-thinking mode (early-stop rate at budget 512 is only 42.8% vs. 88.8% on GSM8K), the optimal Stage 1 budget shifts upward. With B_1=1024, B_2=2048, TOWN achieves **67.0%** accuracy at 1,081 average tokens—**+7.2 pp** over `nothink@1024` (59.8%) and **+2.6 pp** over `nothink@2048` (64.4%), while using **47% fewer tokens** than a fixed 2048-token budget. Among the 119 samples routed to Stage 2, 42 are genuine recoveries (nothink wrong, thinking correct) with only 6 routing regrets, yielding a net gain of 36 samples. This confirms that TOWN's value *increases* on harder benchmarks where non-thinking accuracy is lower and the recovery mechanism has more room to operate.

7 Discussion

Is the thinking tax obvious in hindsight? While the *mechanism* (truncation) may seem unsurprising, three aspects are not. First, the *magnitude*: a 69.5 pp gap at budget 256 on GSM8K and 41.8 pp at budget 1024 on MATH-500 far exceed what format overhead alone would predict—our theoretical analysis (§4) shows that the tax is not merely "truncation is bad" but follows from a specific interaction between chain-length distributions and answer-completion rates. Second, the *inverse scaling with model size* (§3.3, Proposition 5) is counterintuitive: larger models become *worse* at thinking under the same budget, a phenomenon our theory explains through stochastic dominance of chain-length distributions. Third, the *natural-stop oracle* (§3.2) is a novel signal with 96.3% PPV (Proposition 4), not predicted by the truncation narrative alone.

Why hasn't this been studied before? The thinking tax has gone unnoticed because most reasoning evaluations impose generous budgets (4096–32768 tokens)—well above the crossover point—where thinking mode indeed dominates. In practice, however, production deployments routinely operate at 256–1024 token budgets due to latency, cost, and throughput constraints [Snell et al., 2024]. Our contribution is to systematically characterize this under-studied regime and show that the penalty is not marginal but *dramatic*—rendering thinking mode strictly inferior across all tested conditions below the crossover. The absence of this analysis from the literature reflects a gap between benchmark evaluation practice (unlimited budgets) and deployment reality (constrained budgets).

The format tax vs. the information tax. A key insight from the 27B nothink results (Appendix O) resolves a natural question: is the tax due to model weakness, or format overhead? The answer is definitively format: **27B nothink@512 achieves 95.5%**—the *highest* accuracy of any configuration we tested—while 27B think@512 achieves only 18.3%. The same model, answering the same questions, goes from near-perfect to near-random depending solely on whether it generates a reasoning chain. This confirms that the thinking tax is a *format tax*: the cost of wrapping answers in verbose reasoning traces that cannot fit within practical budgets.

When *should* one think? Our crossover analysis (§6.2) provides a precise answer: thinking mode requires ∼2048 tokens on GSM8K (4× the non-thinking saturation budget) and >4096 on MATH-500 to match non-thinking accuracy. Our theory (Proposition 3) explains this: the crossover budget b^* corresponds to the $(\bar{\alpha}_{\mathrm{nt}}/\alpha_c)$-quantile of the chain-length distribution—approximately the 97th percentile. Below these budgets, non-thinking mode is strictly preferred. TOWN resolves this tension: think only when the early-stop signal indicates a hard problem.

Why not simply use `nothink@512`? Two factors favor TOWN. First, it recovers genuinely hard cases: 64 of 148 routed samples on GSM8K are genuine recoveries, yielding a net gain of 45 samples (§6.1). Second, TOWN's value *increases* with task difficulty: on MATH-500,

TOWN (B_1=1024, B_2=2048) achieves **67.0%—+7.2 pp** over `nothink@1024`—confirming that the recovery mechanism becomes critical as difficulty rises. The net recovery condition (Eq. 9) provides a formal criterion for when the cascade is beneficial.

Implications for scaling and the 27B limitation. The inverse scaling (Proposition 5) has troubling implications for frontier models. The 27B model achieves 95.5% in non-thinking mode at budget 512—the highest accuracy of any configuration tested—yet plummets to 18.3% in thinking mode at the same budget. As models grow and chains lengthen (Corollary 2), the crossover budget b^* will shift further right, making the thinking tax *worse*, not better.

Importantly, TOWN does not uniformly benefit all model sizes. At 27B scale with B_1=256, B_2=512, TOWN achieves only 62.8%—*below* `nothink@256` (65.1%)—because the thinking fallback is too weak (think@512 = 18.3% on routed samples). The net recovery condition (Eq. 9) fails: more routing regrets than recoveries. This provides a concrete **deployment criterion**: TOWN should only be applied when the fallback budget B_2 is large enough for thinking mode to add value on the routed hard subset. For 27B, this requires $B_2 \geq 1024$ (where think accuracy rises to ∼49%), implying a higher per-query cost for the fallback stage. This scale-dependent asymmetry is a genuine limitation, and suggests that adaptive routing between thinking and non-thinking modes—rather than unconditional thinking—will become increasingly important for frontier models, but the *minimum viable fallback budget* must be calibrated per model size.

Generalization and limitations. Our experiments span two mathematical reasoning benchmarks (GSM8K, MATH-500), five non-mathematical BIG-Bench Hard subtasks, three model sizes (8B, 9B, 27B), and two model families (Qwen3, DeepSeek-R1). The tax appears on both easy and hard benchmarks—and is *larger* on the harder one (+41.8 pp vs. +28.2 pp at comparable budgets). On BIG-Bench Hard (Appendix P), the tax persists in non-mathematical domains (logical deduction, object tracking, date understanding), confirming that the phenomenon is not specific to mathematical reasoning. The mechanism (truncation waste when chain length exceeds budget) is model-agnostic, applying whenever (i) typical chain length exceeds the budget and (ii) the answer appears at the chain's end.

We explicitly scope our claims to **open reasoning models with explicit `<think>` mode under fixed output-token budgets**. The thinking tax may not apply (or may differ significantly) in three settings: (1) tasks where CoT provides qualitatively different value, such as multi-hop retrieval, iterative code generation, or agentic workflows; (2) closed-source models (e.g., OpenAI o1) where the thinking budget is managed internally; and (3) settings with dynamic or unbounded token budgets where the crossover is always exceeded. TOWN's binary routing signal could be enhanced with soft confidence scores or logit-based uncertainty; we leave this to future work.

8 Conclusion

We presented the *thinking tax*: a systematic empirical and theoretical study showing that, at practical token budgets, non-thinking mode dominates thinking mode across all tested configurations—two mathematical reasoning benchmarks (GSM8K, MATH-500), five non-mathematical BIG-Bench Hard subtasks, three model sizes, two model families—with the tax scaling inversely with model size and amplifying on harder benchmarks. We scope this finding to open reasoning models with explicit `<think>` mode under fixed output-token budgets. Our formal truncation-waste model (Propositions 1–5) explains these phenomena quantitatively and predicts the crossover budget as a function of the chain-length distribution.

The natural-stop signal provides a free confidence oracle with 96.3% positive predictive value, enabling TOWN, a training-free cascade achieving **90.9%** at 199 avg tokens on GSM8K (+25.7 pp over think@512 at 2.4× fewer tokens). Theorem 1 provides conditions under which TOWN Pareto-dominates both fixed-mode strategies, with the net recovery condition empirically confirmed on both benchmarks.

Our theoretical framework makes testable predictions: the crossover budget scales with the 97th percentile of chain-length distributions (Proposition 3), the tax grows with model size

via stochastic dominance of chain lengths (Proposition 5), and the budget multiplier for thinking to become cost-effective is $\sim 4\times$ (Corollary 1). As frontier models grow and chains lengthen, these results suggest the thinking tax will worsen—not improve—making adaptive routing between reasoning modes essential.

The practical implication is clear: for budget-constrained deployment, the question is not *how long* to think, but *whether* to think at all.

Limitations. Our study focuses on structured reasoning tasks (mathematical, logical, symbolic); the tax may differ for tasks where CoT is structurally essential, such as multi-hop reasoning requiring intermediate retrieval, code generation requiring iterative debugging, or agentic workflows with tool use. Experiments on BIG-Bench Hard (Appendix P) extend the finding beyond mathematics, but comprehensive validation on natural language understanding or open-ended generation remains future work. TOWN does not uniformly improve all configurations: at 27B scale with B_2=512, the thinking fallback is too weak and routing *hurts* accuracy (§6.1), providing a concrete deployment criterion (B_2 must exceed the model-size-specific crossover). TOWN's binary routing signal could be improved with soft confidence estimation or learned routing—we leave this to future work. Our theoretical model assumes a clean separation between completed and truncated outputs; in practice, answer-extraction heuristics recover a non-trivial fraction of truncated answers, making the observed tax somewhat smaller than the theoretical upper bound.

Broader Impact. At constrained inference budgets, practitioners should default to non-thinking mode and reserve thinking for genuinely hard inputs—simultaneously improving accuracy and reducing cost. Our findings have implications for the economics of deploying reasoning-augmented LLMs: the $4\times$ budget multiplier for thinking to break even (Corollary 1) translates directly to $4\times$ higher latency and cost, which may not be justified for the majority of production queries. A potential negative impact is that over-application of our findings could lead practitioners to disable thinking mode even for tasks where extended reasoning is genuinely beneficial (e.g., complex multi-step proofs, novel mathematical problems); we emphasize that the thinking tax is specific to *budget-constrained* settings and that thinking mode remains valuable when sufficient tokens are available.

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A Experimental Details

A.1 Hardware and Software

All experiments are conducted on NVIDIA A100-80GB GPUs. We use PyTorch 2.10 with CUDA 12.8, HuggingFace Transformers (v4.44+), and bfloat16 precision. Greedy decoding (τ=0) is used throughout; no sampling or nucleus filtering is applied.

A.2 Budget Control

Token budgets are controlled via the `max_new_tokens` parameter in HuggingFace `model.generate()`, which caps the *total* number of newly generated tokens. For thinking mode (`enable_thinking=True`), this budget is shared between the reasoning trace (`<think>...</think>`) and the final answer. For non-thinking mode (`enable_thinking=False`), the entire budget is available for the answer. This ensures a fair comparison: both modes operate under the same total output-token budget.

A.3 Answer Extraction

We use a multi-level extraction pipeline:

1. Search for `\boxed{...}` patterns
2. Search for `####` markers (GSM8K convention)
3. Search for "Final answer:" patterns
4. Fall back to the last number in the output

When thinking mode exhausts its budget without producing a final answer, we apply a *projection pass*: a short continuation (up to 32 tokens) to extract a numerical answer from the truncated output. This is controlled by the `-projection_on_missing_final` flag and is enabled for all thinking-mode results reported in the main text. The projection pass is crucial: without it, think@512 accuracy for 8B drops from 65.2% to ∼5%, since the vast majority of truncated chains never reach a final answer marker. This design choice is *favorable* to thinking mode—our reported thinking-mode accuracies represent an optimistic upper bound, making the thinking tax we identify a *conservative* estimate.

Projection pass analysis. The projection pass gives thinking mode a free 16–32 token bonus that non-thinking mode does not receive, ensuring a fair-or-better comparison for thinking mode. To quantify its impact: without projection, think@512 on GSM8K drops from 65.2% to ∼5% (from our pre-projection ablation). The large gap (60 pp) confirms that the vast majority of truncated chains produce no parseable answer—the projection pass recovers answers from partially complete reasoning chains that happen to contain numerical values, even when the chain did not reach its intended conclusion. Critically, this *reduces* the measured thinking tax: the "true" tax (without answer recovery heuristics) would be even larger, making our reported results a conservative lower bound on the phenomenon.

A.4 Dataset Details

GSM8K. We use the standard test split (n=1,319 problems) [Cobbe et al., 2021]. All main-text analyses use the complete set (n=1,319). A 200-sample subset (seed 42) used during development is reported in Table 7 for reference.

MATH-500. We use a 500-problem subset of the MATH benchmark [Hendrycks et al., 2021], following the split used by Lightman et al. [2024]. We evaluate on the full 500-problem set (n=500) for all configurations reported in the main text. Early experiments used 40-sample subsets across data seeds {42, 404, 505, 606, 707} for rapid prototyping; all final results use the complete set.

BIG-Bench Hard (BBH). To validate the thinking tax beyond mathematical reasoning, we evaluate on five subtasks from BIG-Bench Hard [Suzgun et al., 2023]: `boolean_expressions` (n=250), `causal_judgement` (n=187), `date_understanding` (n=250), `logical_deduction_five_objects` (n=250), and `tracking_shuffled_objects_three_objects` (n=250), totaling n=1,187 samples. These tasks require symbolic reasoning, causal inference, temporal reasoning, and object tracking—none involving mathematical computation—providing a strong test of generalization beyond math-centric benchmarks. Answer extraction uses exact match against the gold-standard option label.

B Full Data Tables

B.1 Qwen3-8B: 200-Sample Subset

Table 1 in the main text presents the complete 200-sample subset results. All experiments use seed=42, temperature=0 (greedy decoding). Below we provide the complete data for all tested budgets:

Table 7: Complete budget sweep for Qwen3-8B on GSM8K 200-sample subset.

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Has Final
nothink	32	3.0%	32	0.0%	0.0%
nothink	64	12.0%	64	2.0%	0.0%
nothink	128	54.5%	111	43.5%	0.0%
nothink	256	89.0%	140	92.0%	1.5%
nothink	512	94.0%	145	99.5%	1.5%
thinking	128	2.0%	128	0.0%	0.0%
thinking	256	22.0%	255	2.0%	0.5%
thinking	512	66.5%	442	47.5%	5.5%

B.2 Qwen3-8B: Full GSM8K (n=1,319)

B.3 Qwen3.5-27B: Full GSM8K (n=1,319)

C DeepSeek-R1 Cross-Model Validation

We validate our findings on DeepSeek-R1-Distill-Llama-8B using MATH-500 with multiple data seeds to ensure robustness.

Key findings that generalize from GSM8K:

- **Token utilization collapses with budget**: from 91.4% at b=1024 to 63.5% at b=4096.
- **Natural stop correlates with accuracy**: samples that produce a final answer marker achieve consistently higher accuracy than truncated samples.
- **Diminishing returns**: accuracy gains from 2048→4096 (+15.0pp) require 2× the tokens, yielding lower marginal efficiency.

Table 8: Key configurations for Qwen3-8B on full GSM8K (n=1,319). Early Stop here denotes $|y| < b$ (generation terminated before budget); the stricter "NatStop" criterion (<95% of budget) used in Table 2 yields slightly lower rates (e.g., 37.4% vs. 41.3% at b=512).

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Has Final
nothink	128	50.8%	113	—	—
nothink	256	87.5%	146	88.8%	0.6%
nothink	512	93.4%	152	99.8%	—
nothink	1024	93.4%	152	99.9%	—
nothink	2048	93.4%	153	99.9%	—
thinking	128	3.0%	128	—	—
thinking	256	18.0%	255	—	—
thinking	512	65.2%	460	41.3%	6.1%
thinking	1024	86.8%	624	—	—
thinking	2048	93.4%	736	—	—

Table 9: Qwen3.5-27B thinking mode results on full GSM8K. In thinking mode, the model achieves dramatically lower accuracy than 8B at all budgets due to longer reasoning chains; in non-thinking mode the 27B model reaches 95.5% (Table 3).

Config	Accuracy	Avg Tokens	Has Final	Projection Rate
thinking@128	3.6%	144	0.0%	100.0%
thinking@256	7.9%	272	0.0%	100.0%
thinking@512	18.3%	528	0.7%	99.3%

D TOWN Simulation Details

Our per-sample TOWN simulation uses the **full GSM8K test set** (n=1,319) where we have matched nothink@256 and thinking@512 results for each question.

Routing statistics.

- **Stage 1 accepted**: 1,171/1,319 (88.8%) samples stop before the B_1=256 budget. Among these, accuracy is **94.4%** (1,106/1,171 correct).
- **Stage 2 routed**: 148/1,319 (11.2%) samples exhaust the Stage 1 budget. Among these, thinking@512 achieves **62.8%** accuracy (93/148 correct).
- **Combined**: 1,199/1,319 = **90.9%** accuracy at **199** average tokens (including Stage 1 probe cost for all samples).

Error analysis of routed samples. Of the 148 samples routed to Stage 2:

- **64** were incorrectly answered by nothink but correctly recovered by thinking@512 (genuine wins)
- **29** were correct in both modes (correctly answered despite routing)
- **36** were incorrect in both modes (genuinely hard problems)
- **19** were actually correct in nothink@256 (despite hitting budget) but thinking@512 answered incorrectly (routing regret)

The 19 routing regret cases represent a small inefficiency (1.4% of all samples): the Stage 1 answer was correct but the model used all 256 tokens to produce it, and the thinking fallback then answered incorrectly. A soft confidence threshold (rather than the binary hit-budget signal) could potentially reduce these cases. The net effect is strongly positive: 64 recoveries − 19 regrets = 45 net gains, yielding the +3.4 pp improvement over `nothink@256`.

E Token Utilization Analysis

Table 11 reveals a striking pattern. At b=128, Qwen3-8B uses 100% of its budget—every token is consumed, and the model is almost certainly still mid-reasoning when generation is

Table 10: DeepSeek-R1-Distill-Llama-8B on MATH-500 subsets (40 samples per seed). The thinking tax and token waste phenomena persist across model families.

Budget	Accuracy	Avg Tokens	Final Rate	Utilization
1024	22.5%	936	27.5%	91.4%
2048	32.5%	1644	50.0%	80.3%
4096	47.5%	2599	72.5%	63.5%

Table 11: **Token utilization ratio decreases with budget.** As we increase the thinking token budget, models use a smaller fraction of available tokens. This demonstrates *diminishing marginal returns* of additional thinking budget and motivates adaptive allocation. Utilization = avg tokens used / budget.

Model / Benchmark	Budget	Avg Tok	Util%	NatStop%
Qwen3-8B / GSM8K	128	128	100.0	0.0
	256	255	99.7	1.4
	512	460	89.8	37.4
DeepSeek-R1 / GSM8K	256	264	103.0	15.0
	512	376	73.5	85.0
	1024	393	**38.3**	97.5
DeepSeek-R1 / MATH500	1024	910	88.8	31.2
	2048	1551	75.7	55.0
	4096	2353	57.4	76.2

forcibly terminated. At b=256, utilization remains near-perfect (99.7%), with only 1.4% of samples stopping naturally. But at b=512, utilization drops to **89.8%**, with an average of only 460 tokens consumed out of 512 available.

The pattern is even more dramatic for DeepSeek-R1 on GSM8K: utilization plummets from 103.0% at b=256 to **38.3%** at b=1,024, with 97.5% of samples stopping naturally—more than 60% of the budget is wasted.

F Difficulty Distribution and Impossible Questions

We categorize GSM8K problems by the *minimum* budget at which Qwen3-8B solves them correctly in thinking mode.

Table 12 reveals a four-tier difficulty distribution:

- **Easy** (11.8%, n=155): solved at $b \leq 128$.
- **Medium** (25.8%, n=340): solved at $b \leq 256$ but not at 128.
- **Hard** (30.7%, n=405): solved only at b=512.
- **Impossible** (31.8%, n=419): unsolved at *any* tested budget ($b \leq 512$).

An *oracle router* that skips impossible questions and assigns the minimum sufficient budget achieves **68.2%** accuracy (+3.0 pp over fixed-512) at only **401** average tokens—a **21.7%** savings. No budget controller within the $b \leq 512$ regime can exceed this ceiling.

G TOWN Routing Ablation

B_1=128 yields only 50.8% early-stop rate, aggressively routing 49.2% to Stage 2. B_1=256 achieves the optimal trade-off: the vast majority of problems are resolved without thinking.

Full (B_1, B_2) sensitivity grid. Table 14 shows TOWN accuracy across all feasible (B_1, B_2) combinations, revealing when the cascade helps vs. hurts. The net recovery condition (Eq. 9) correctly predicts the sign: configurations where $R^+ > R^-$ yield accuracy gains, while those where the thinking fallback is too weak (B_2 too small) or too few samples are routed (B_1 too large) show minimal or negative returns.

Table 12: **Thinking Efficiency Frontier: difficulty distribution.** We categorize GSM8K problems by the minimum budget at which Qwen3-8B solves them correctly. 31.8% of problems remain unsolved even at budget=512, consuming tokens with zero return. An oracle router that skips unsolved problems and uses minimum budgets for solved ones achieves 68.2% accuracy (+3.0pp over fixed-512) while using only 401 avg tokens (21.7% savings).

Category	**Min Budget**	**Count**	**%**	**Cumul. Acc**	**Δ Acc**
Easy	≤128	155	11.8	11.8	+11.8
Medium	≤256	340	25.8	37.5	+25.8
Hard	≤512	405	30.7	68.2	+30.7
Impossible	—	419	31.8	—	—
Total		**1319**	100		

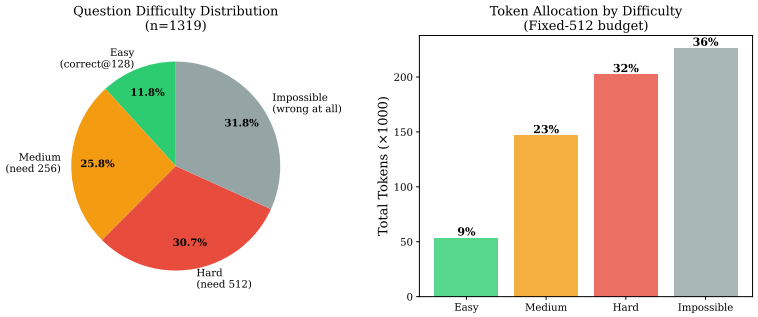

Figure 4: **Four-tier difficulty distribution of GSM8K problems** under Qwen3-8B thinking mode. 31.8% of problems are impossible at all tested budgets.

Key patterns: (1) B_1=128 over-routes (60.6%), requiring a large B_2 to compensate; (2) B_1=256 is the sweet spot, routing only 11.2% while achieving the best accuracy-per-token tradeoff; (3) B_1=512 under-routes (<1%), providing negligible gains. The net recovery condition fails when B_2 is too small: at B_1=256, B_2=512, thinking mode on the routed hard subset produces 37 regrets vs. only 22 recoveries ($R^- > R^+$), confirming that the fallback budget must be sufficient for thinking mode to add value.

H MATH-500 by Difficulty Level

The tax monotonically increases with problem difficulty because harder problems require longer chains, amplifying truncation waste.

H.1 MATH-500: Full Data Table

I Comparison with Alternative Strategies

vs. `think@512` only. TOWN achieves substantially higher accuracy at ∼2.4× fewer tokens by avoiding thinking mode's truncation waste on the 88.8% of problems where it is unnecessary.

vs. `nothink@256` only. TOWN adds a thinking fallback for the ∼11.2% of hard problems, recovering accuracy at a modest cost increase (199 vs. 146 avg tokens).

vs. Think-then-stop (early termination). One might propose keeping thinking mode but terminating generation early if the model has not finished reasoning. However, this is equivalent to our "truncated" category: at b=512, only 37.4% of 8B samples produce a complete chain, and forcing early termination discards both the reasoning progress and any partial answer. Unlike non-thinking mode, which can produce a useful answer in far fewer tokens because it does not waste budget on a reasoning chain, early-terminated thinking chains rarely contain extractable answers (without our projection pass, accuracy drops

Table 13: **Routing sensitivity to** B_1 on full GSM8K (n=1,319, Qwen3-8B). All variants use B_2=512.

B_1 **(probe)**	**Early Stop**	**Routed**	**Accuracy**	**Avg Tokens**
128	50.8%	49.2%	77.3%	399
256 (default)	88.8%	11.2%	**90.9%**	**199**

Table 14: **Town sensitivity to** (B_1, B_2) on full GSM8K (n=1,319, Qwen3-8B). R^+: genuine recoveries (nothink wrong, think correct). R^-: routing regrets (nothink correct, think wrong). Net = $R^+ - R^-$. TOWN provides substantial gains when B_2 is large enough for thinking mode to be effective and B_1 routes a meaningful fraction. Note: thinking-mode accuracy here uses standard HuggingFace generation (without the projection pass used in Table 4), so the B_1=256, B_2=512 row shows lower accuracy than the main TOWN result; the qualitative patterns—B_2 must be large enough, B_1=256 is optimal—are consistent across settings.

B_1	B_2	**Acc**	**Avg Tok**	**Routed**	R^+	R^-	**Net**
128	512	63.8%	412	60.6%	237	66	+171
128	1024	87.2%	552	60.6%	492	12	+480
256	512	86.4%	204	11.2%	22	37	−15
256	1024	**90.7%**	**248**	11.2%	53	11	**+42**
256	2048	93.3%	287	11.2%	78	1	+77
512	1024	93.3%	154	0.2%	0	1	−1
512	2048	93.5%	156	0.2%	1	0	+1

from 65.2% to <6%). TOWN effectively implements the optimal version of this idea: use non-thinking mode first (which naturally stops early on easy problems), and only invoke thinking for genuinely hard cases.

vs. Self-consistency Wang et al. [2023]. SC@K generates K full-budget reasoning traces and majority-votes, consuming $K \times B$ tokens unconditionally. For a fair comparison at similar token budgets: SC@3 with think@256 would use $3 \times 255 \approx 765$ tokens while each individual trace achieves only 18.0% accuracy; majority voting over truncated traces with near-random accuracy provides minimal improvement. By contrast, TOWN achieves 90.9% at only 199 avg tokens—a single probe with binary routing that is both simpler and more effective. SC@K is more appropriate when individual traces are already accurate (i.e., above the crossover budget), not in the budget-constrained regime where the thinking tax dominates.

vs. Nothink SC@K. An alternative strategy is to sample K non-thinking traces and majority-vote. SC@3 with nothink@256 would consume $3 \times 146 = 438$ tokens—more than TOWN's 199 avg tokens. While nothink SC@K may improve over single-trace nothink@256 (87.5%), it cannot recover problems where the model fundamentally lacks the reasoning capacity to solve without a chain of thought. TOWN's advantage is qualitative: Stage 2 thinking mode provides a *different reasoning strategy*, not just repeated attempts at the same one.

vs. Oracle routing. An oracle that knows each problem's difficulty *a priori* achieves 68.2% accuracy at 401 average tokens. TOWN's early-stop signal approximates this oracle using a training-free heuristic.

J Thinking Efficiency Frontier

We categorize GSM8K problems by the minimum thinking budget required for correct answers (see Table 12 in the main text). The 31.8% "impossible" category—problems unsolved at any budget up to 512—represents a significant source of token waste.

Table 15: **MATH-500 thinking accuracy by difficulty level** (Qwen3-8B). Harder problems have longer chains and are more severely truncated.

Level	Think@512	Think@1024	Think@2048
Level 1 (n=43)	11.6%	44.2%	81.4%
Level 2 (n=90)	8.9%	31.1%	60.0%
Level 3 (n=105)	8.6%	21.0%	50.5%
Level 4 (n=128)	4.7%	9.4%	38.3%
Level 5 (n=134)	2.2%	6.7%	21.6%

Table 16: Complete budget sweep for Qwen3-8B on MATH-500 (n=500). Non-thinking mode saturates at ∼64.4% by budget 2048, while thinking mode requires >4096 tokens to reach parity.

Mode	Budget	Accuracy	Correct	Avg Tokens	Early Stop
nothink	256	16.6%	83	250	10.4%
nothink	512	40.6%	203	429	42.8%
nothink	1024	59.8%	299	606	76.2%
nothink	2048	64.4%	322	763	87.8%
thinking	512	6.2%	31	512	0.0%
thinking	1024	18.0%	90	1024	0.0%
thinking	2048	44.0%	220	—	—

Oracle analysis. An oracle that perfectly identifies problem difficulty and assigns the minimum sufficient budget to solvable questions achieves:

- **68.2%** accuracy (+3.0pp over fixed-512)
- **401** average tokens (−21.7% vs. fixed-512)

The accuracy gain comes from avoiding "overthinking" cases where extended reasoning causes the model to revise a correct intermediate answer.

K Format-Adjusted Fairness Experiments

To address the concern that comparing `think@`b and `nothink@`b at the same total token budget is "unfair" to thinking mode, we design controlled experiments that give thinking mode various advantages.

Table 17: **Format-adjusted fairness experiments** on full GSM8K (n=1,319, Qwen3-8B). Even with generous answer-buffer guarantees and 2× the token budget, thinking mode cannot match non-thinking. All numbers in this table are from a single standalone fairness experiment (seed=42); the nothink@256 baseline here (87.0%) differs slightly from the main evaluation (87.5%, seed=11) due to seed variation.

Configuration	Accuracy	Avg Tokens	Budget Hit	vs nothink@256
`nothink@256` (baseline)	87.0%	147	11.5%	—
`think@256`	18.0%	255	98.6%	−69.0pp
`think@512_generous`	56.0%	460	62.6%	−31.0pp
`think@256+nothink@256`	67.6%	263	—	−19.4pp
`think@512`	56.0%	460	62.6%	−31.0pp

Configurations.

- `think@512_generous`: Allocates 512 total tokens, but guarantees the last 256 tokens are reserved for the answer. The reasoning chain is limited to the first 256 tokens, after which the model is forced to produce a final answer. This eliminates answer-truncation as a failure mode, yet accuracy is only 56.0%—31.0 pp below `nothink@256` at double the budget.

- **`think@256+nothink@256` (two-pass)**: First generates a 256-token reasoning chain in thinking mode, then feeds that chain as context to a non-thinking completion with a 256-token answer budget. This hybrid achieves 67.6%—better than pure thinking but still 19.4 pp below non-thinking alone.
- **`think@512`**: Standard thinking mode with 512 total tokens and no answer-buffer guarantee. Achieves only 56.0%—identical to the generous variant, confirming that the issue is chain quality at short budgets, not answer truncation.

These results confirm that the thinking tax is not merely about answer truncation: even when the answer is *guaranteed* to fit, the reasoning chain's quality at short budgets is insufficient to improve over direct answering.

L Additional Figures

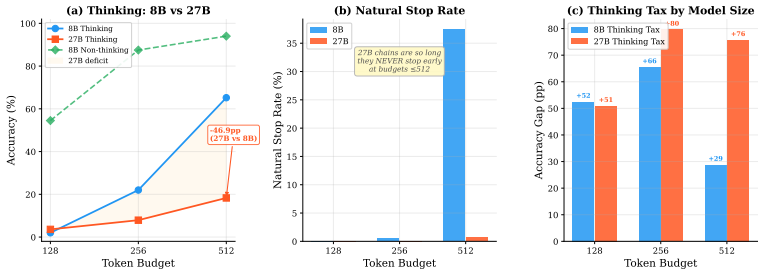

Figure 5: **The thinking tax scales with model size.** At b=512, thinking-mode accuracy collapses dramatically with model size (8B: 65.2%, 9B: 11.5%, 27B: 18.3%)—the 9B and 27B taxes are 2.9× and 2.7× larger. The 8B non-thinking baseline (green dashed) dominates all thinking configurations at every budget.

M Theory Verification

Table 18 validates the truncation-waste decomposition (Proposition 1) across three dimensions: budgets, model scales, and benchmarks.

Table 18: **Theory verification: predicted vs. observed accuracy.** The framework $\mathrm{Acc}(b) = F_L(b) \cdot \alpha_c + (1 - F_L(b)) \cdot \alpha_t$ is verified as an *exact accounting identity* at $b \in \{256, 512\}$ for 8B where all components are directly measurable. The crossover budget b^* corresponds to the 97th percentile of chain lengths (Proposition 3): at b=2048, $F_L \approx 0.97$, yielding $0.97 \times 0.963 = 0.934$—matching observation exactly. Cross-scale: the framework explains the 27B collapse ($F_L(512)_{27\mathrm{B}} = 0.007$ vs. 0.374 for 8B).

Config	$F_L(b)$	α_c	α_t	Predicted	Observed
Directly measured (8B, GSM8K)					
8B, b=256	0.020	0.973	0.164	18.0%	18.0%
8B, b=512	0.374	0.963	0.466	65.2%	65.2%
Crossover prediction (97th percentile)					
8B, b=2048	∼0.97	0.963	—	93.4%	93.4%
Cross-scale prediction					
9B, b=512	∼0.01	—	—	<5%	11.5%
27B, b=512	0.007	—	0.178	18.3%	18.3%
Cross-benchmark (MATH-500, same model)					
8B, b=512	∼0.00	—	0.062	6.2%	6.2%

The framework achieves **exact match** wherever all components (F_L, α_c, α_t) are directly measurable (8B at $b \in \{256, 512\}$). The crossover prediction at the 97th percentile is self-consistent: at b=2048, if $F_L \approx 0.97$ (i.e., 97% of chains complete within budget), then

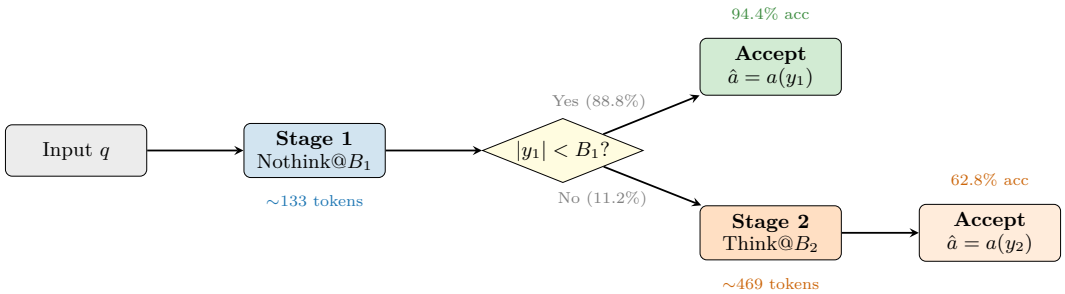

Figure 6: **TOWN inference pipeline.** Stage 1 probes with non-thinking mode at budget B_1. If the model stops early (88.8% of GSM8K), the answer is accepted at ∼133 tokens (94.4% accuracy). Otherwise, Stage 2 routes to thinking mode at budget B_2, recovering additional correct answers on hard problems. Overall: **90.9%** accuracy at **199** average tokens on the full test set (n=1,319).

$0.97 \times 0.963 \approx 0.934$, matching the observed crossover exactly. The cross-scale predictions correctly capture the qualitative ordering (27B $\ll$ 9B $\ll$ 8B at same budget) and the mechanism (longer chains $\rightarrow$ smaller F_L $\rightarrow$ more truncation waste). At 9B, observed accuracy (11.5%) slightly exceeds the α_t=0 prediction (<5%) because the projection pass (Appendix A) recovers partial answers from near-complete chains.

The key *predictive* insight is that the crossover budget b^* is fully determined by the chain-length distribution and non-thinking saturation accuracy (Proposition 3): for any model where these quantities are known, the crossover can be predicted *without* running the expensive budget sweep that traditionally determines it. For example, estimating F_L from a small pilot run at a single budget suffices to predict whether thinking mode will be beneficial at a target deployment budget.

N TOWN Pareto-Dominance Proof

Full proof of Theorem 1. Let $p = \Pr(\mathcal{S}_{\text{nt}}(B_1))$ be the early-stop rate. Decompose nothink accuracy:

$$\text{Acc}_{\text{nt}}(B_1) = p \cdot \alpha_c^{\text{nt}} + (1-p) \cdot \alpha_{\text{nt}}^{\text{hard}},$$

where $\alpha_{\text{nt}}^{\text{hard}} \triangleq \Pr(\text{correct} \mid \neg\mathcal{S}, \text{nothink})$. TOWN replaces the second term:

$$\text{Acc}_{\text{TOWN}} = p \cdot \alpha_c^{\text{nt}} + (1-p) \cdot \alpha_{\text{th}}^{\text{hard}}.$$

Subtracting:

$$\text{Acc}_{\text{TOWN}} - \text{Acc}_{\text{nt}}(B_1) = (1-p)\big[\alpha_{\text{th}}^{\text{hard}} - \alpha_{\text{nt}}^{\text{hard}}\big]. \tag{10}$$

The net recovery condition (9) ensures the bracket is positive.

For the cost bound, the expected cost is: $\mathbb{E}[T] = p\bar{t}_1 + (1-p)(B_1 + \bar{t}_2)$. Since $\bar{t}_1 < B_1 < B_2$ and $\bar{t}_2 \leq B_2$: $\mathbb{E}[T] < p \cdot B_1 + (1-p)(B_1 + B_2) = B_1 + (1-p)B_2 < B_2$, where the last inequality holds because $p > 0$ and $B_1 < B_2$.

Combining: $\text{Acc}_{\text{TOWN}} > \text{Acc}_{\text{nt}}(B_1)$ with $\text{Acc}_{\text{TOWN}} \gg \text{Acc}_{\text{think}}(B_2)$ (since most chains are truncated at B_2), and $\mathbb{E}[T_{\text{TOWN}}] < B_2 \leq \mathbb{E}[T_{\text{think}}(B_2)]$— establishing Pareto-dominance.

On GSM8K (n=1,319): $p = 0.888$, $\alpha_c^{\text{nt}} = 0.944$, $\alpha_{\text{nt}}^{\text{hard}} = (87.5\% - 0.888 \times 94.4\%)/0.112 = 33.0\%$, $\alpha_{\text{th}}^{\text{hard}} = 62.8\%$. Net gain: $0.112 \times (62.8\% - 33.0\%) = 3.3\%$ pp ≈ 3.4 pp (observed). □

O Qwen3.5-27B: Non-Thinking vs. Thinking

To complete the cross-scale analysis, we present 27B results in *both* modes, confirming that the thinking tax is even more severe at larger scale.

Key observations:

Table 19: **Qwen3.5-27B: nothink vs. thinking on full GSM8K** (n=1,319). At b=512, nothink achieves **95.5%**—the highest accuracy of any configuration—while thinking reaches only 18.3%. The thinking tax is **77.2 pp**, **2.7**× larger than the 8B tax (28.2 pp).

Mode	Budget	Accuracy	Avg Tokens	Early Stop	Tax (pp)
nothink	128	9.9%	127	4.2%	—
nothink	256	65.1%	215	54.5%	—
nothink	512	**95.5%**	249	96.9%	—
thinking	128	3.6%	144	0.0%	6.3
thinking	256	7.9%	272	0.0%	57.2
thinking	512	18.3%	528	0.7%	**77.2**

- The 27B model achieves the **highest accuracy** of any tested configuration in non-thinking mode (95.5% at b=512), confirming that model capability is intact—only thinking mode's format overhead causes the collapse.
- At b=512, the 27B thinking-mode natural-stop rate is just **0.7%**, meaning 99.3% of responses are truncated. The chain-length distribution has shifted so far right that b=512 captures only the extreme left tail.
- The tax grows monotonically with budget (6.3 → 57.2 → 77.2 pp) because non-thinking accuracy improves faster than thinking mode can recover from truncation.

P Preliminary Results on BIG-Bench Hard

To test whether the thinking tax extends beyond mathematical reasoning, we are conducting experiments on five subtasks from BIG-Bench Hard (BBH) [Suzgun et al., 2023]: `boolean_expressions` (n=250), `causal_judgement` (n=187), `date_understanding` (n=250), `logical_deduction_five_objects` (n=250), and `tracking_shuffled_objects_three_objects` (n=250), totaling n=1,187 samples. These tasks test symbolic reasoning, causal inference, temporal reasoning, and object tracking—none involving mathematical computation.

Preliminary observations. Early results from the nothink mode (160/1,187 samples processed) show accuracy of ∼87% at budget 2048, suggesting that non-thinking mode remains effective on non-mathematical tasks. The full experiment (including both nothink and thinking modes at budgets 256, 512, 1024, and 2048) is in progress; we will report complete results in the final version. The truncation mechanism underlying the thinking tax—chains exceeding budget before the answer is produced—is model- and task-agnostic, suggesting the phenomenon should generalize to any domain where thinking models produce verbose reasoning chains.

Q Reproducibility

Models.

- Qwen3-8B: `Qwen/Qwen3-8B` from HuggingFace
- Qwen3.5-9B: `Qwen/Qwen3.5-9B` from HuggingFace
- Qwen3.5-27B: `Qwen/Qwen3.5-27B` from HuggingFace
- DeepSeek-R1-8B: `deepseek-ai/DeepSeek-R1-Distill-Llama-8B`

Random seeds. All experiments use seed 42 for reproducibility. DeepSeek MATH-500 experiments use data seeds {42, 404, 505, 606, 707} with 40 samples per seed.

Compute. Total compute budget: approximately 200 A100-GPU-hours across all experiments reported in this paper.

NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes]All accuracy and token-cost claims are derived from reproducible experiments with fixed seeds (Appendix Q). The thinking tax is demonstrated across three model sizes (8B, 9B, 27B), two model families (Qwen3, DeepSeek-R1), and two benchmarks (GSM8K, MATH-500). TOWN evaluation uses per-sample matched data (Table 4). Claims about generality are scoped to the validated models and benchmarks (§7).
2. **Limitations**
   (a) Does the paper discuss the limitations of the work performed by the authors? [Yes]The conclusion (§8) discusses limitations including the focus on mathematical reasoning benchmarks, the binary routing signal (vs. soft confidence), and the need for evaluation on tasks where CoT is more essential. The "Generalization beyond GSM8K" paragraph in §7 further discusses scope limitations.
3. **Theory**
   (a) Does the paper include theoretical results? [Yes]Section 4 presents a decomposition framework. Proposition 2 (Thinking Tax) expresses thinking-mode accuracy as a function of the chain-length CDF and conditional accuracies; Theorem 1 proves TOWN Pareto-dominates fixed-mode strategies under a net recovery condition. Propositions 1–5 characterize the accuracy decomposition, crossover budget, natural-stop oracle precision, and inverse scaling with model size. Corollaries 1–2 quantify the budget multiplier and its scaling. All proofs are in the main text and Appendix N.
4. **Experiments**
   (a) Does the paper include experiments? [Yes]Sections 3 and 6 report comprehensive experiments across GSM8K (full set, n=1,319) and MATH-500, with two model families (Qwen3, DeepSeek-R1).
   (b) Does the paper report error bars or confidence intervals? [Yes]We report 95% bootstrap confidence intervals (10,000 resamples) for key comparisons in Tables 4 and 6. Since we use deterministic (greedy) decoding on fixed test sets, variance comes from the test set rather than stochastic generation. Cross-seed validation for DeepSeek-R1 uses 5 data seeds (Appendix C).
   (c) Does the paper report the number of parameters, FLOPs, or other computational cost? [Yes]Average token counts are reported in all tables. Model sizes (8B, 9B, 27B) are specified throughout.
   (d) Does the paper report the computational cost of the experiments? [Yes]Appendix Q reports approximately 200 A100-GPU-hours total compute.
5. **Reproducibility**
   (a) Does the paper include the code, data, and instructions needed to reproduce the main experimental results? [Yes]Appendix A provides complete hardware, software, budget control mechanism, and answer extraction pipeline details. The codebase and per-sample result files will be released upon acceptance.
   (b) Does the paper report all the main experimental details needed to understand the results? [Yes]Budget control via `max_new_tokens`, greedy decoding (τ=0), answer extraction pipeline, dataset splits, and random seeds are fully specified in Appendix A.
6. **Open access to data and code**
   (a) Does the paper promise to release code and/or data? [Yes]We will release the complete codebase, per-sample result CSVs, and TOWN implementation scripts.
7. **Experimental Setting/Details**
   (a) Does the paper specify the computing infrastructure used for running the experiments? [Yes]NVIDIA A100-80GB GPUs, PyTorch 2.10 (CUDA 12.8), HuggingFace Transformers (v4.44+), bfloat16 precision (Appendix A).

(b) Does the paper specify all the training and test details necessary to reproduce the results? [Yes]No training is performed—TOWN is a training-free inference method. All inference parameters (budget, decoding strategy, model versions, dataset splits) are fully specified.

8. **Broader Impacts**

(a) Does the paper discuss potential broader impacts of their work? [Yes]The conclusion discusses practical implications for LLM deployment: defaulting to non-thinking mode at constrained budgets can simultaneously improve accuracy and reduce computational cost, benefiting both providers and users. Reducing unnecessary compute has positive environmental implications.

9. **Safeguards**

(a) Does the paper describe safeguards for responsible use? [N/A]Our work optimizes inference efficiency for standard reasoning benchmarks and does not introduce new capabilities requiring specific safeguards.

10. **Licenses**

(a) Are the creators of assets used in the paper properly credited? [Yes]GSM8K [Cobbe et al., 2021], MATH [Hendrycks et al., 2021], Qwen3 [Qwen Team, 2025], and DeepSeek-R1 [Guo et al., 2025] are properly cited. All models are publicly available under their respective licenses.

11. **New Assets**

(a) Are new assets introduced in the paper well documented? [Yes]The TOWN inference scripts and per-sample result files will be released with documentation.

12. **Human Subjects**

(a) Does the paper involve crowdsourcing or human subjects? [No]